Project Proposal

Department of Computer Engineering

Course Name: Big Data Analytics
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TrustGauge - Amazon Reviews Analyzer

Project Abstract

This project aims to develop a web application that analyzes customer reviews from the Amazon US Customer Reviews. The application will allow users to search for a product and receive a confidence score based on customer sentiment analysis. The sentiment scores will be derived from NLP processing of customer reviews, providing insights into individual products and their parent companies. This tool aims to assist consumers in making informed purchasing decisions based on aggregated sentiment analysis.

Problem Statement

With the vast number of products available on ecommerce platforms like Amazon, it is challenging for consumers to gauge the overall sentiment and reliability of a product solely from individual reviews. Many reviews may be biased, inconsistent, or manipulated. A system that can aggregate and analyze reviews to provide a reliable confidence score based on sentiment analysis will help customers make data-driven decisions, improving their shopping experience.

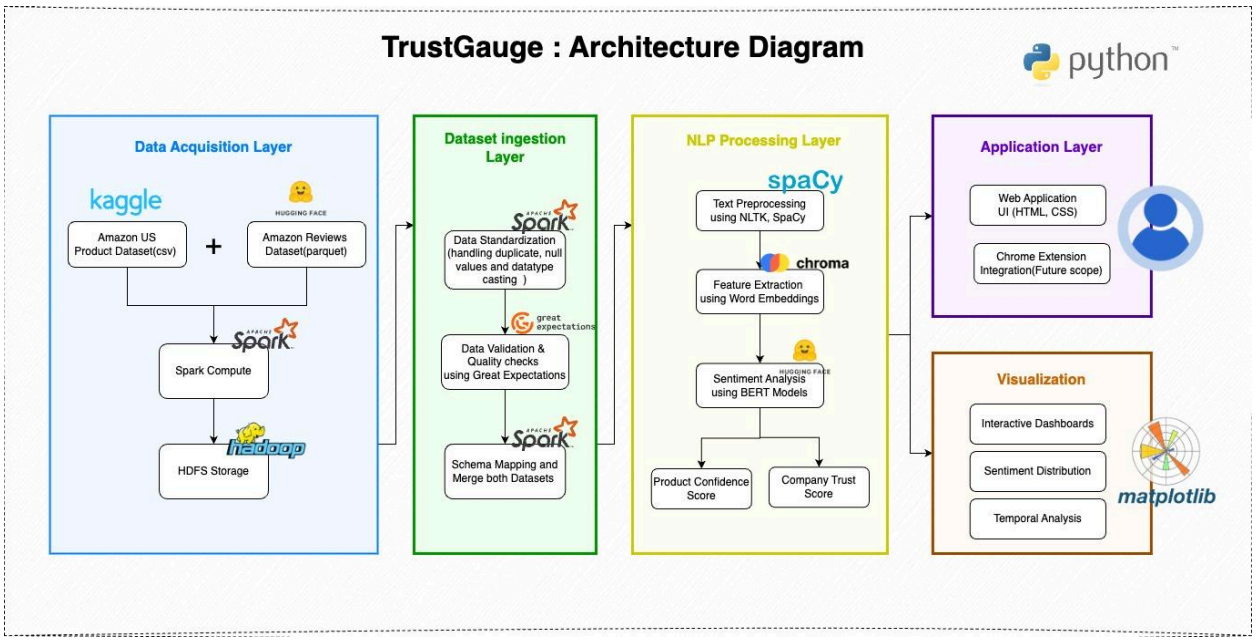
Objectives

- Develop a web application that allows users to search for products and receive a confidence score based on sentiment analysis of customer reviews.
- Implement NLP techniques to process and analyze large volumes of textual data efficiently.
- Provide insights into individual products and their parent companies based on sentiment trends.

- Ensure scalability and performance by leveraging Apache Spark for big data processing.
- Design an intuitive user interface using Gradio to make sentiment analysis results easily accessible to consumers.
- Enhance customer decision-making by aggregating and analyzing reviews to mitigate bias and inconsistencies.

Architecture

TrustGauge is designed to analyse and measure trustworthiness in product reviews by leveraging Big Data Analytics and Natural Language Processing techniques. Our approach leverages multiple datasets, data acquisition, ingestion, processing, visualization and applies sentiment analysis to generate meaningful insights.Below is the architecture diagram to show flow of data through different layers.



We'll be using Apache PySpark for distributed computing throughout our Big Data Ingestion Pipeline, spaCy and BERT models for NLP and interactive dashboards for data representation. We ensure a comprehensive and reliable trust evaluation system. Future enhancements include Chrome extension integration for easy accessibility for trust score. TrustGauge is built with the vision of empowering consumers with data-driven trust insights.

Data Source

Data Source 1 Name: Amazon US Customer Reviews Dataset

Data Source 1 Link: Kaggle Dataset

Data File Size: Approximately 3.63 GB

Approximate Number of Records: *6,922,459*

Data Source 2 Name: Amazon Reviews 2023

Data Source 2 Link: Github Dataset

Data File Size: Approximately 113 GB

Approximate Number of Records: *512 Million*

Features Breakdown: The dataset comprises customer reviews from the Amazon.com marketplace and relevant metadata from 1995 to 2015. It is designed to support research on the characteristics and evolution of customer reviews, including how individuals assess and articulate their experiences with products on a large scale.

Proposed Technologies and Programming Languages

- **Python:** Primary language for data processing and NLP.
- **Apache Spark:** To handle large-scale data processing efficiently.
- **NLTK** (Natural Language Toolkit): For text preprocessing and sentiment analysis.
- **Numpy & Pandas**: For numerical computing and data manipulation.
- **Google Colab:** For collaborative development and model training.
- **Gradio:** To create an interactive web-based UI.
- **HTML & CSS:** For frontend development.

Future Scope

The future development of this project aims to transition from a web application to a **Chrome extension**, allowing users to view sentiment analysis directly on Amazon product pages. This extension will enhance user experience by providing real-time sentiment scores alongside actual customer reviews. Additionally, improvements in review accuracy will be made by refining sentiment analysis techniques to better align with authentic customer feedback, ensuring more reliable and data-driven purchasing decisions.